Given conditions :

Coal production per year = 5×10^6 Ton

Coal seam thickness = 10 m

S.R. = $3 m^3/ton$

Overburden thickness = S.R. x coal seam thickness x ρ_{coal}

$= 3 \times 10 \times 1.6$

$= 48\ m$

4 Benches of 12m each

Assumptions :-

		coal	overburden
(i)	Drill diameter	150 mm	250 mm
(ii)	Burden	4.5	6
(iii)	spacing	6.5	8
(iv)	Bench height	10	12
(v)	Length of drill rod	6	7
(vi)	Time to set drill m/c (min)	6	7
(vii)	Time to set drill rod (min)	3	3
(viii)	Rate of drilling (min/m)	2	4
(ix)	Subgrade drilling	0	10 %
(x)	overall utilisation	0.6	0.6
(xi)	Hours per year	3000	3000

Calculations :

volume of coal handling / year $= \frac{5 \times 10^6}{1.6} = 3125000\ m^3$

vol. of coal handled per hole $= 4.5 \times 6 \times 10$

$= 292.5\ m^3$

No. of holes req/yr $= \frac{3125000}{292.5} = 10683.76$

Actual no. of holes req./yr = 10684

No. of feed req. $= \frac{10}{6} = 1.66$

Actual no. of feed req. $= 2$

Total Time to drill one hole $= 6 + 3 + 10 \times 2$

$= 29$ min

No. of holes drilled per hour $= \frac{60}{29} = 2.069$

Actual no. of holes / year $= 3000 \times 2.069 \times 0.6$

$= 3724.2$

No. of drills $= \frac{10684}{3724.2} = 2.86$

Actual no. of drills $= 3.$

Vol. of OB $= 3 \times 5 \times 10^3 = 15 \times 10^3 \ m^3$

vol. of OB handling / hole $= 6 \times 8 \times 12$

$= 576$

No. of holes req. $= \frac{15 \times 10^3}{576} = 26041.667$

Actual no. of holes req. $= 26042$

Drilling bench height with subgrade drilling $= 13.2$ m

No. of feed req. $= \frac{13.2}{7} \simeq 2$

Total time $= 7 + 3 + 4 \times 13.2$

$= 62.8$ min

Actual holes/hr $= \frac{60}{62.8} = 0.955$

Actual holes/year $= 3000 \times 0.955 \times 0.6$

$= 1719.25$

$$\text{No. of drill req.} = \frac{26042}{1719.25} = 15.14$$

$\therefore$ Actual no. of drill req. $= 16.$